



Customer Behavior Analysis in E-Commerce

Data Mining Study of UK Online Retail Transactions

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Abstract

This project analyzes transactional data from a UK-based online retail store. Our goal is to reveal insights into customer purchasing behavior. The dataset includes over 500,000 transactions from December 2010 to December 2011. We applied data mining techniques including data cleaning, association rule mining with the FP-Growth algorithm, K-means clustering, and time-series analysis. Key findings include the identification of frequent product pairings, distinct customer segments based on spending, and seasonal spikes in sales volume. These findings support strategies for targeted marketing, inventory planning, and customer engagement.

Introduction

As the e-commerce industry continues to expand, so does the opportunity to leverage transactional data for strategic analysis. This project examines a dataset from a UK-based online gift retailer and applies data mining techniques to find patterns that address key business questions. The primary objective is to analyze customer purchasing behavior in order to support data-driven decision-making across five key areas: 1) customer segmentation based on spending, 2) seasonal and temporal purchase trends, 3) product associations for cross-selling opportunities, 4) holiday season performance of specific product categories, 5) geographic trends in customer demographics. By applying clustering, association rule mining, and time-series analysis, this project demonstrates how structured data analysis can provide more effective retail strategies and enable personalized shopping experiences.

Data Source and Description

The dataset used in this project is the "Online Retail" dataset, obtained from the UCI Machine Learning Repository. It contains transactional data of an online retail store based in the UK, spanning a period from December 1, 2010, to December 9, 2011. An example of the raw transactional data is shown in table 1, which includes customer-specific and product-level details extracted from the original dataset.

The dataset comprises the following key features:

- **InvoiceNo:** A unique identifier for each transaction.
- **StockCode:** A unique code for each product.
- **Description:** A textual description of the product.
- **Quantity:** The number of units of a product purchased in a transaction.
- **InvoiceDate:** The date and time of the transaction.
- **UnitPrice:** The price of a single unit of the product.
- **CustomerID:** A unique identifier for each customer.
- **Country:** The country where the customer resides.

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Date	Time
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	2010-12-01	08:26:00
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	2010-12-01	08:26:00
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	2010-12-01	08:26:00
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	2010-12-01	08:26:00

Table 1: Sample of transactional data showing product and customer attributes in the original dataset.

Data Quality and Preprocessing

The Online Retail dataset, while a valuable source of information, presented certain data quality challenges that required careful preprocessing. The following steps were taken to address these issues and ensure data integrity:

1. **Handling Missing Values:** The dataset contained a significant number of missing values, particularly in the CustomerID column. To address this, rows with missing CustomerID values were removed from the dataset to ensure that only transactions with identifiable customer information were considered for analysis. Figure 1 shows the distribution of missing values across the dataset prior to cleaning.

```

RangeIndex: 541909 entries, 0 to 541908
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   InvoiceNo        541909 non-null object
1   StockCode       541909 non-null object
2   Description     540455 non-null object
3   Quantity       541909 non-null int64
4   InvoiceDate     541909 non-null datetime64[ns]
5   UnitPrice      541909 non-null float64
6   CustomerID     406829 non-null float64
7   Country        541909 non-null object
8   Date           541909 non-null object
9   Time           541909 non-null object

```

Figure 1: Missing value distribution across data set features before cleaning.

2. **Removing Canceled Orders:** To ensure data accuracy, transactions representing canceled orders were identified by the presence of "C" in the InvoiceNo column. These canceled orders were removed from the dataset to prevent distortions in purchase patterns

and trends. Notably, some customers exhibited a significantly higher cancellation rate, with certain individuals canceling over 200 orders, indicating a potential area for further investigation. Figure 2 highlights the top 10 customers with the highest number of cancellations.

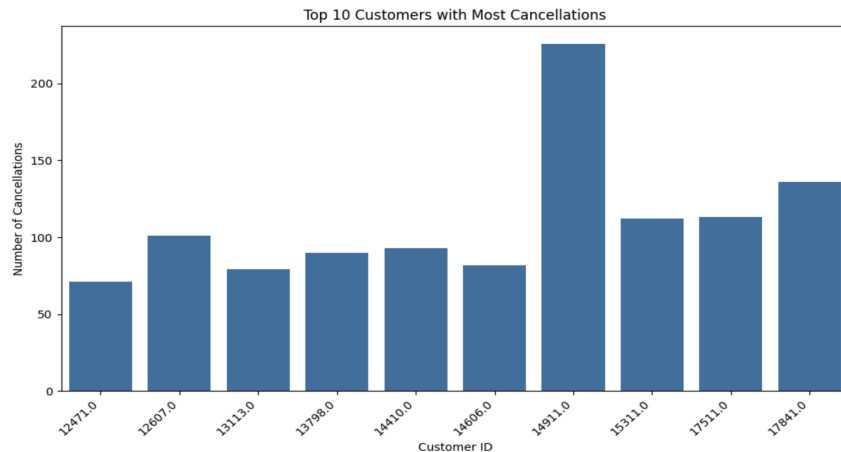


Figure 2: Top 10 customers by number of canceled orders.

3. **Filtering Irrelevant Stock Codes:** Stock codes not starting with a number were deemed irrelevant for the analysis and were filtered out. This ensured that only valid product identifiers were considered. Figure 3 shows the proportion of stock codes before and after filtering.

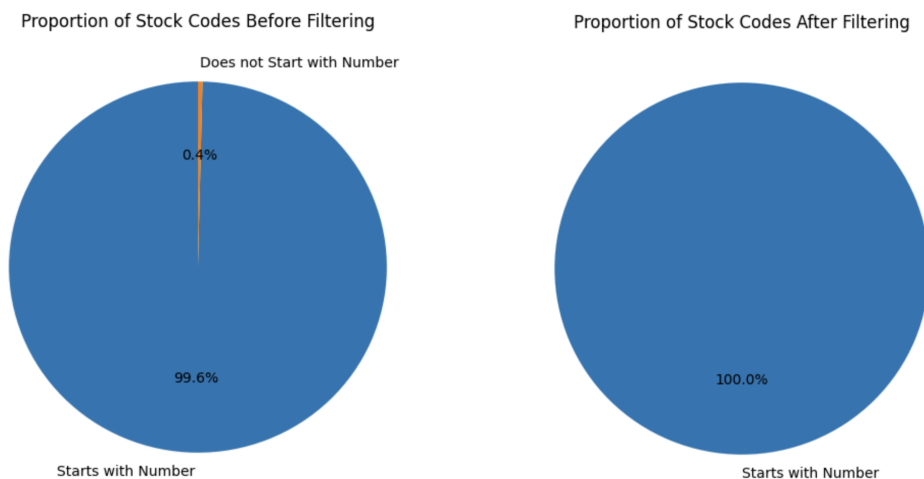


Figure 3: Comparison of stock code proportions before and after filtering.

4. **Handling Outliers:** Outliers in our numerical variables such as 'Total Spent per InvoiceNo', 'Total', 'Quantity', and 'UnitPrice' were identified using the Interquartile Range (IQR) method. Outliers were replaced with upper and lower bounds calculated based on the IQR to minimize their influence on the analysis. Box and whisker plots were used to better understand that each variable follows a similar distribution. Figures 4 and 5 show the distributions of these variables before capping.

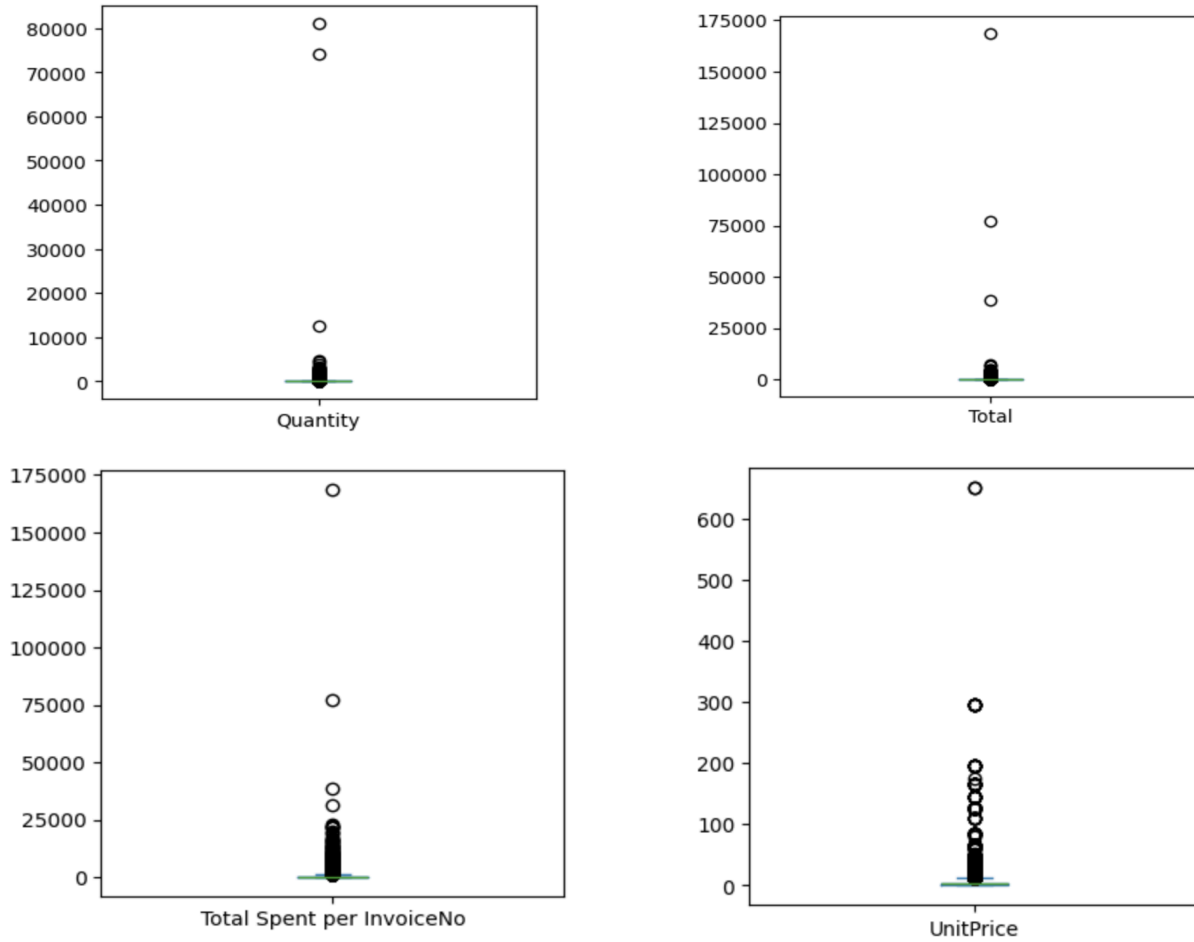
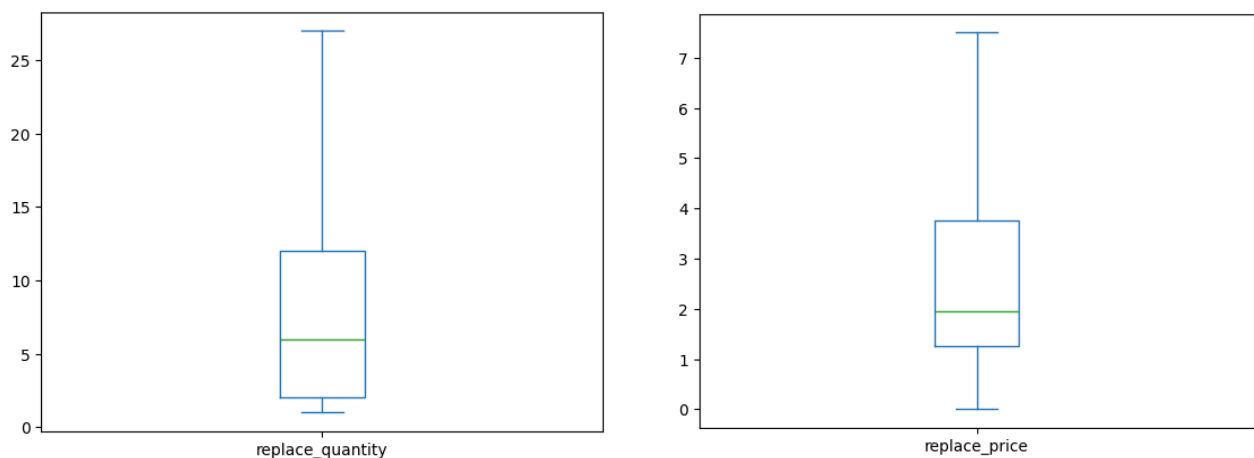


Figure 4: Boxplots showing outliers in Quantity, Total, Total Spent per InvoiceNo, and UnitPrice



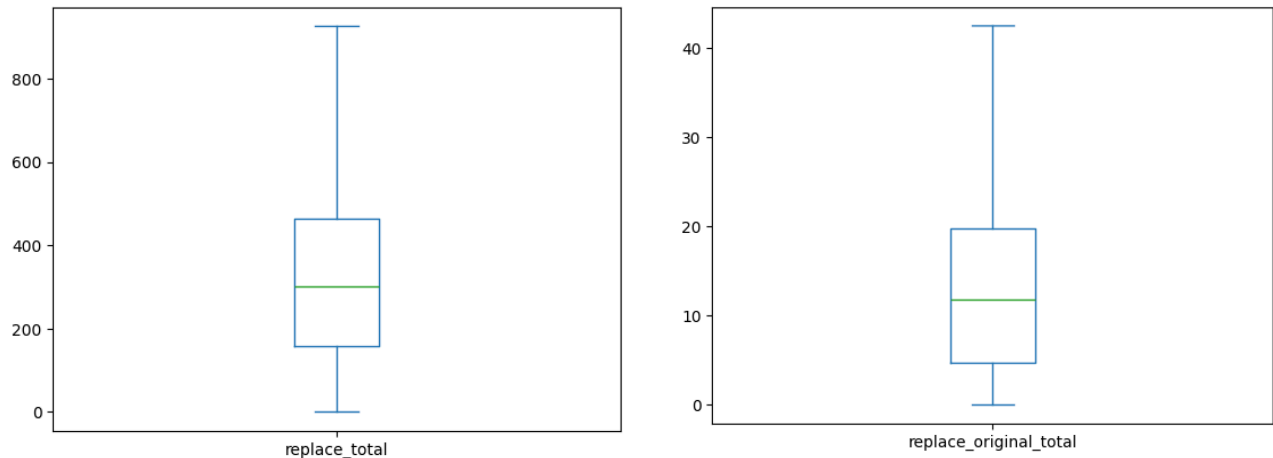


Figure 5: Boxplots without outliers for `replace_total`, `replace_price`, `replace_quantity`, and `replace_original_total`

5. **Data Transformation:** The `InvoiceDate` column was converted to datetime objects using `pd.to_datetime` to enable extraction of date and time components, such as month, day, and hour. This facilitated the analysis of temporal trends and seasonal patterns in purchasing behavior. Additionally, by plotting weekly purchase trends, it was revealed that the business was closed on Saturdays, as indicated by the absence of transactions on that day. This pattern along with peak activity on Thursdays, is shown in Figure 6 and is crucial for understanding the overall sales patterns and customer behavior.

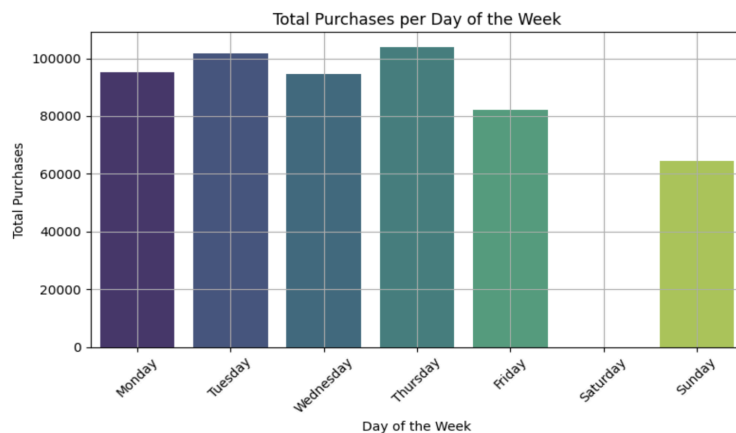


Figure 6: Total purchases per day of the week, showing peak activity on Thursdays and no purchases on Saturdays.

Feature Engineering

To facilitate a more comprehensive analysis, new features were engineered from the existing data:

1. **Date and Time Features:** The InvoiceDate column was utilized to extract 'Month', 'Date', and 'Time' features. These features facilitate analysis of seasonal trends and temporal patterns in purchasing behavior.
2. **Total Transaction Value:** A new column named 'Total' was added to a replica of our original data titled df_ver002. It was created by multiplying the 'Quantity' and 'UnitPrice' columns. This feature represents the total value of each transaction, enabling analysis of spending patterns and individual product consumption. The resulting table structure is shown in Table 2.

	InvoiceNo	StockCode	Description	Quantity	UnitPrice	CustomerID	Country	Date	Time	Total
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2.55	17850.0	United Kingdom	2010-12-01	08:26:00	15.30
1	536365	71053	WHITE METAL LANTERN	6	3.39	17850.0	United Kingdom	2010-12-01	08:26:00	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2.75	17850.0	United Kingdom	2010-12-01	08:26:00	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	3.39	17850.0	United Kingdom	2010-12-01	08:26:00	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	3.39	17850.0	United Kingdom	2010-12-01	08:26:00	20.34

Table 2: Sample structure of the enhanced dataset (df_ver002) with calculated Total column

3. **Total Spent per Invoice:** 'Total Spent per InvoiceNo' was calculated by grouping transactions by 'CustomerID' and 'InvoiceNo' and summing the 'Total' values. This feature aggregates spending per invoice and, to do this, exists in a new table (df_ver003) that removes the StockCode, Quantity, and Unit Price columns. Table 3 highlights analysis of customer purchase behavior at the invoice level that can be used to better understand how customers invest into the business.

	CustomerID	InvoiceNo	Date	Time	Total Spent per InvoiceNo
0	12346.0	541431	2011-01-18	10:01:00	77183.60
1	12347.0	537626	2010-12-07	14:57:00	711.79
2	12347.0	542237	2011-01-26	14:30:00	475.39
3	12347.0	549222	2011-04-07	10:43:00	636.25
4	12347.0	556201	2011-06-09	13:01:00	382.52

Table 3: Aggregated view of customer spending per invoice (df_ver003) based on CustomerID and InvoiceNo.

4. **Category and DayOfWeek:** Using GPT-3.5 Turbo API, we were able to organize our products into categories for further analysis. Additionally, the day of the week in which each product was purchased was added to better analyze spending patterns throughout the week. These two columns exist in a new data table, df_ver006.

	InvoiceNo	StockCode	Description	CustomerID	Country	Date	Time	replace_original_total	replace_quantity	replace_price	Total Spent per InvoiceNo	0	Category	DayOfWeek
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	17850.0	United Kingdom	2010-12-01	08:26:00	15.30	6.0	2.55	15.30	WHITE HANGING HEART T-LIGHT HOLDER	Decor	Wednesday
1	536365	71053	WHITE METAL LANTERN	17850.0	United Kingdom	2010-12-01	08:26:00	20.34	6.0	3.39	20.34	WHITE METAL LANTERN	['Lighting', 'item']	Wednesday
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	17850.0	United Kingdom	2010-12-01	08:26:00	22.00	8.0	2.75	22.00	CREAM CUPID HEARTS COAT HANGER	['Home', 'Decor']	Wednesday
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	17850.0	United Kingdom	2010-12-01	08:26:00	20.34	6.0	3.39	20.34	KNITTED UNION FLAG HOT WATER BOTTLE	Wellness	Wednesday
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	17850.0	United Kingdom	2010-12-01	08:26:00	20.34	6.0	3.39	20.34	RED WOOLLY HOTTIE WHITE HEART.	['warmth', 'comfort']	Wednesday

Table 4: Sample of df_ver006 containing the new Category and DayOfWeek categories.

Feature Selection

After data cleaning and feature engineering, feature selection was performed to focus on the most relevant variables for customer segmentation and trend analysis. Correlation analysis and domain knowledge were used to identify features strongly associated with customer spending behavior and product preferences. Based on this evaluation, the following features were selected for further analysis:

- **CustomerID:** To identify unique customers and track their individual purchase patterns.
- **TotalSpent:** To represent the overall spending of each customer, serving as a key feature for customer segmentation.
- **Country:** To analyze the geographical distribution of customers and understand regional variations in purchasing behavior.
- **Month:** To uncover seasonal purchase trends and identify peak and off-seasons for product demand.
- **Category:** To explore product preferences and identify top-selling categories during different periods.
- **Description:** To analyze the specific products purchased and understand customer preferences at a granular level.
- **StockCode:** To identify individual products and analyze their purchase frequency.

- **Quantity:** To understand the volume of products purchased in each transaction.
- **UnitPrice:** To analyze the price points of products and their relationship with customer behavior.

This detailed description of the data quality, preprocessing, feature engineering, and feature selection processes ensures transparency and reproducibility of the analysis. It also provides a solid foundation for applying data mining techniques in the subsequent sections of the report.

EDA

Customer Location Analysis

To explore geographic trends in customer behavior, the cleaned dataset was grouped by country based on customer billing locations. The United Kingdom had the largest number of customers in the dataset, which largely impacted the scale of visualizations.

To address this, two figures are provided: Figure 7 displays the full distribution of customer counts by country, including the UK. Figure 8 excludes the UK, allowing for clearer comparison of customer presence across other countries.

When the UK was excluded, Germany, France, and Spain had the highest customer counts.

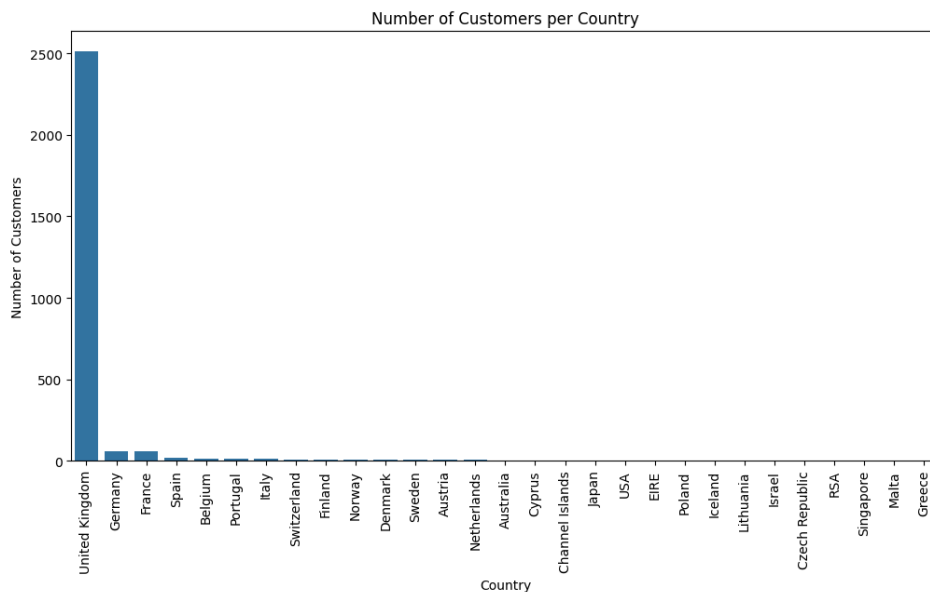


Figure 7: Customer distribution by country, including the United Kingdom.

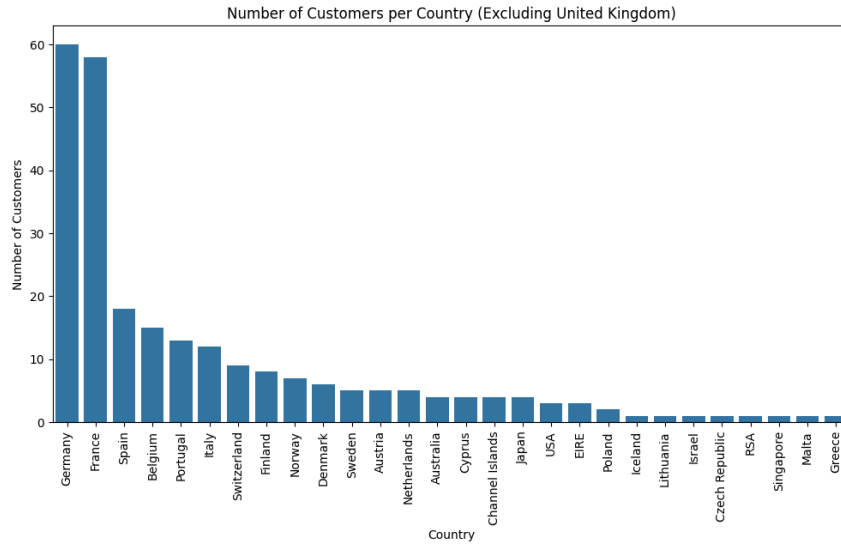


Figure 8: Customer distribution by country, excluding the United Kingdom to portray visibility of other countries.

Purchasing Trends

To explore when customers are most active and what they tend to buy, transaction patterns across time and product categories were examined. This section highlights monthly purchasing behavior and noteworthy product preferences during the holiday season.

As shown in Figure 9, transaction volume steadily increases from mid-year with a major spike beginning in September and peaking in November. This trend suggests that Q4 (October-December) accounts for a large portion of annual purchases.

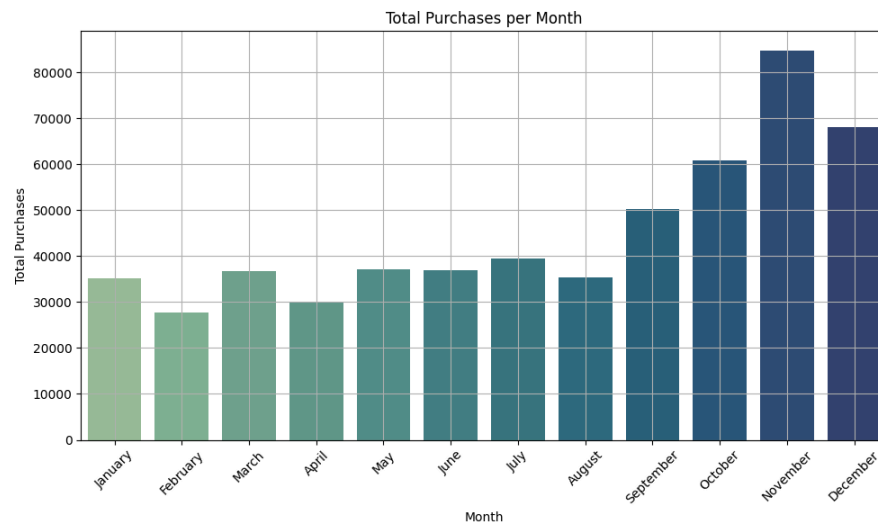


Figure 9: Total purchases per month, with a peak in November reflecting seasonal demand.

Further analysis of holiday season purchases by product category (see Figure 10) illustrates that decor and wellness items were the most frequently purchased during this period. Then followed by lighting products, and serveware/stands, which, while less popular still observed increased interest.

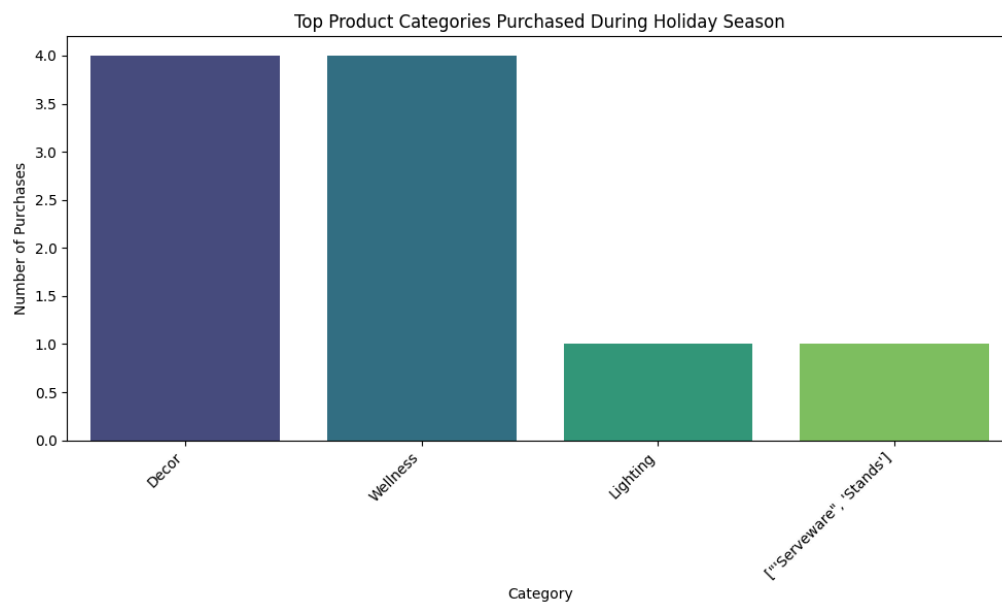


Figure 10: Top product categories purchased during the holiday season, which is led by decor and wellness items.

Methods

Customer Segmentation

To identify the patterns in customer spending behavior, K-means clustering was applied.. The objective was to group customers into distinct segments based on how much they spent during the period of observation. This was performed on the cleaned dataset, referred to in our code as df_ver0002.

Feature Used: TotalSpent, calculated as Quantity X UnitPrice, which was aggregated by each unique CustomerID.

Data Cleaning: Transactions with zero or negative total spend were removed to avoid skewing to the clustering.

Normalization: The TotalSpent variable was normalized using StandardScaler to minimize the impact of extreme values and ensure equal weighting.

K-Value Selection: K=3 was chosen due to it being a reasonable value to reveal low, medium, and high-spending customers. This was used based on business logic to keep results interpretable when used for marketing.

Process: 1) Aggregated total spending per customer, 2) Removed invalid or zero-spend customers, 3) Scaled the data using StandardScaler, 4) Labeled the resulting clusters as: **cluster 0** “low spenders”, **cluster 1** “high value VIP customers”, **cluster 2** “mid-tier spenders”.

The segmentation results are shown in Figure 11.

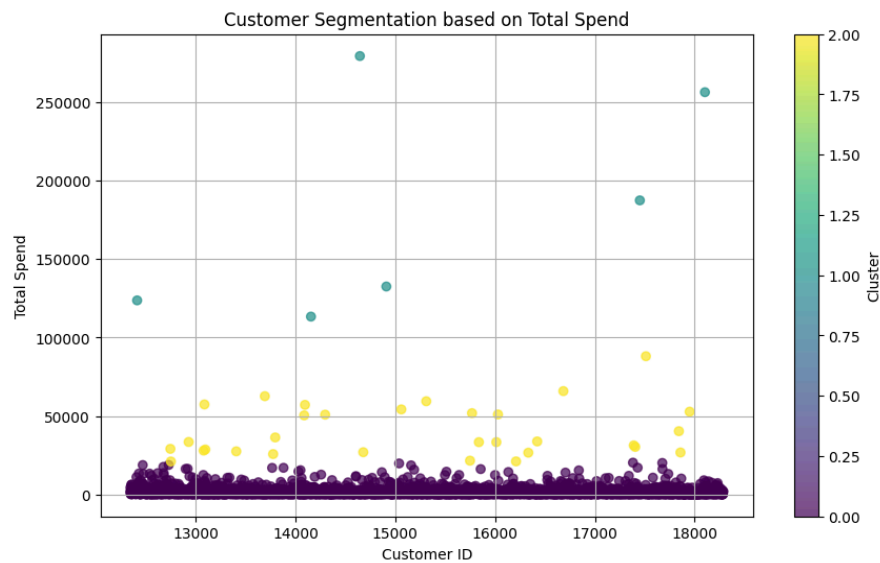


Figure 11: Customer segmentation using clustering based on total spending. Each color represents a distinct cluster of spending behavior.

To further understand spending behavior of each cluster, the average total spend per customer was calculated and visualized. As shown in Figure 12, Cluster 1 (High-Value VIP Customers) had the highest average spend, followed by Cluster 2 (Mid-tier Spenders), while Cluster 0 (Low Spenders) had the lowest average spend.

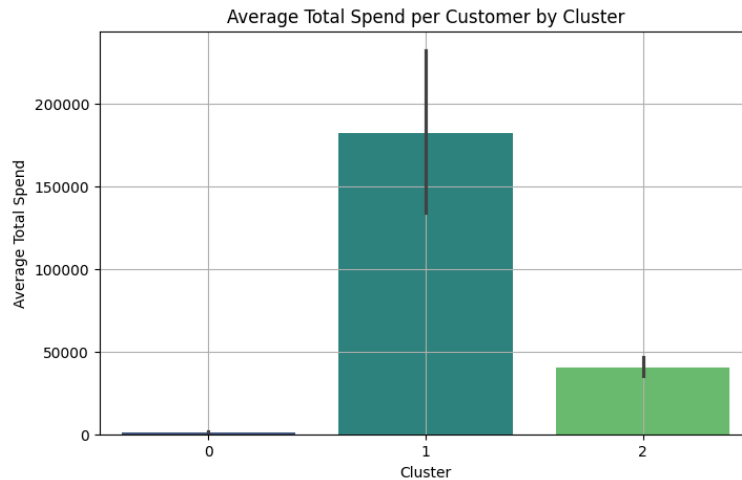


Figure 12: Average total spend per customer across clusters. Cluster 1 reveals significantly higher spending behavior compared to Clusters 0 and 2.

The summary in Table 5 describes the statistical details for each customer cluster including count, average spending, and spending range. Cluster 0, the largest group, includes 4,296 customers with an average total spend of approximately \$1,412. Cluster 1, although the smallest, includes 6 customers, represents high-value buyers whose average spend exceeds 193,000. Cluster 2, consists of 32 mid-tier customers, averaging around \$47,850 in total spending.

	count	mean	min	max
Cluster				
0	4296	1412.279469	3.75	21429.39
1	6	193776.476667	124564.53	279138.02
2	32	47851.724687	25301.54	116729.63

Table 5: Summary statistics of customer clusters showing count, mean, minimum, and maximum total spend per cluster.

Association Rule Mining

To find patterns in product co-purchases and support strategic decision-making such as cross-selling, we applied association rule mining using the FP-Growth algorithm. Although the data was prepared in a format suitable for Apriori through a binary customer-product matrix, FP-Growth was selected due to its superior computational efficiency on large datasets.

Approach:

The dataset was transformed into a binary matrix where rows represented unique CustomerIDs and columns represented individual StockCodes. Each cell was assigned a value of 1 if the customer had purchased the product, and 0 otherwise. This structure mirrors the format used in Apriori-based market basket analysis. We used the FP-Growth algorithm from the mlxtend library to extract frequent itemsets without the overhead of candidate generation. The rule generation process was guided by the following thresholds:

- **Minimum support:** 0.01 (products must appear in at least 1% of customer records)
- **Minimum lift:** 1.0 (to make sure that item pairs appear together more often than would be expected by chance)
- **Confidence:** evaluated as a rule quality metric but not used as a filter

Results Summary:

The algorithm produced over 100 high-quality rules, each consisting of an antecedent (item(s) bought) and a consequent (item(s) likely to be bought). Rules were evaluated based on support, confidence, and lift. Figure 13 presents a scatterplot where each point represents a rule. Rules are plotted by support (x-axis) and confidence (y-axis) with lift visualization through point size and color. The visual helps identify rules that are not only common but also reliable predictors of associated purchases.

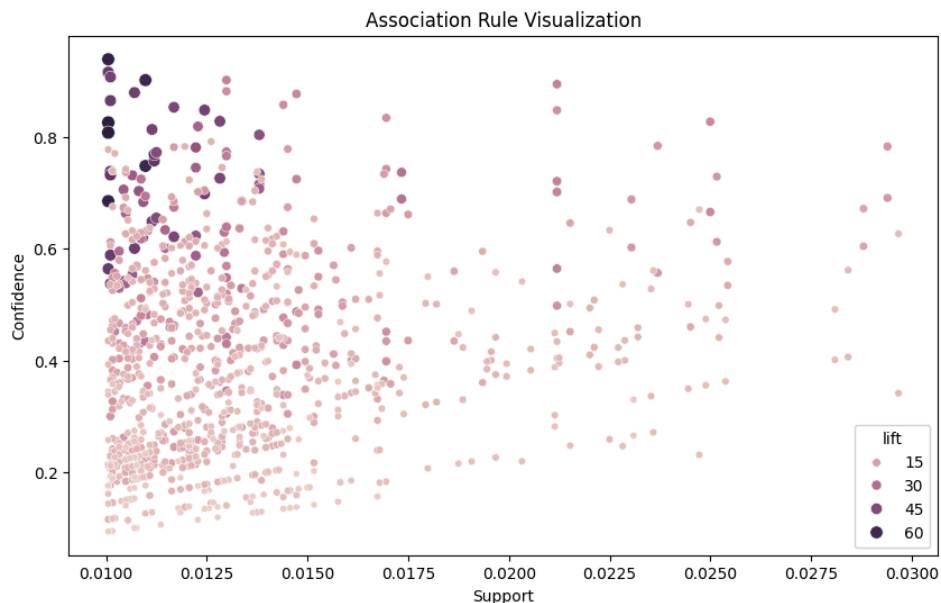


Figure 13: Scatterplot of association rules by support and confidence, with lift represented by point size and color.

To further highlight specific high-performing rules, Figure 14 shows a heatmap of lift values for the top 7 associations. Notably, the Regency Tea Plate series had the highest co-purchase affinities, with lift values exceeding 66. Another significant pattern emerged between Poppy's Playhouse Bedroom and Poppy's Playhouse Kitchen, suggesting that related children's items may perform better when promoted together. By identifying high-lift co-purchase patterns, this analysis reveals strong associative relationships that can inform further study of customer behavior and product performance.

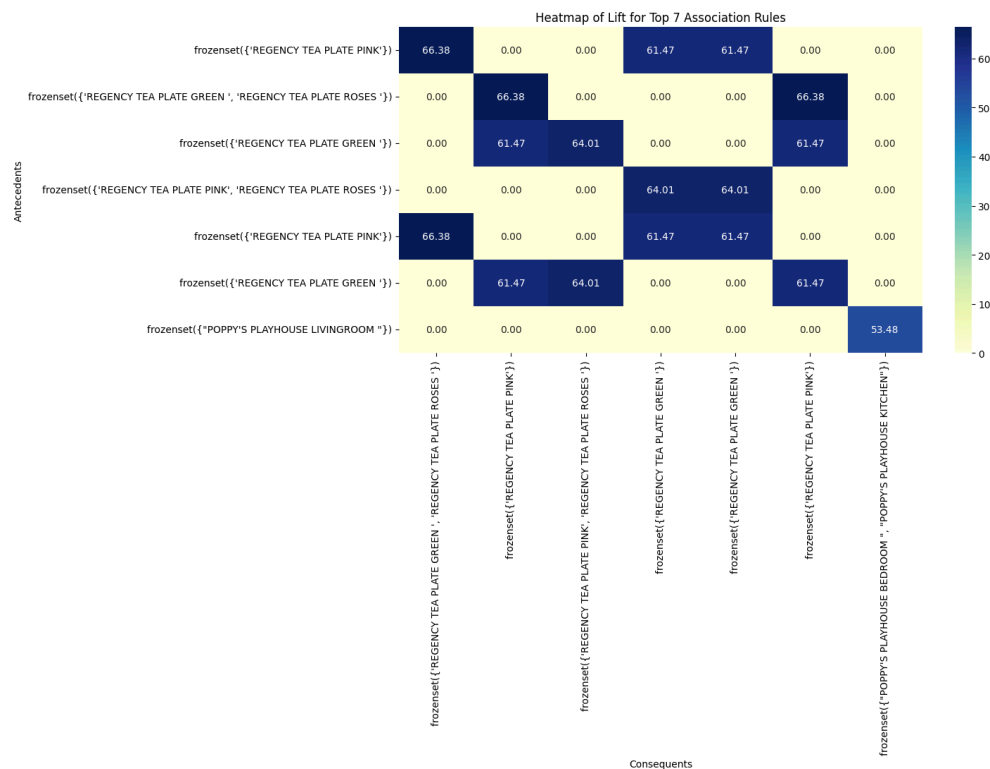


Figure 14: Heatmap of lift values for the top 7 product association rules.

Product Prediction/Recommendation:

Based on the products previously purchased by a customer, additional product recommendations can be generated using association rule mining. The `recommend_product` function analyzes the customer's purchase history and compares it with identified patterns of items frequently bought together. For Customer 16535, for instance, the function suggested a curated list of complementary items such as teacups, tea plates, and Christmas-themed decorations. These

recommendations are not random; they are derived from established associations between products, ensuring relevance and increasing the likelihood of customer interest and satisfaction.

```
Recommendations for customer 16535:  
1. ROSES REGENCY TEACUP AND SAUCER  
2. REGENCY TEA PLATE ROSES  
3. GREEN REGENCY TEACUP AND SAUCER  
4. WOODEN STAR CHRISTMAS SCANDINAVIAN  
5. SET OF 3 CAKE TINS PANTRY DESIGN  
6. PARTY BUNTING  
7. WHITE HANGING HEART T-LIGHT HOLDER  
8. PINK REGENCY TEACUP AND SAUCER  
9. REGENCY CAKESTAND 3 TIER  
10. REGENCY TEA PLATE GREEN  
11. REGENCY TEA PLATE PINK  
12. WOODEN TREE CHRISTMAS SCANDINAVIAN  
13. REGENCY TEAPOT ROSES  
14. ASSORTED COLOUR BIRD ORNAMENT
```

Figure 15: Example output for recommended products for a customer.

Conclusion

By analyzing over half a million transactions from a UK-based online retailer, we found clear patterns in customer behavior, purchase timing, and product relationships. Our customer segmentation approach revealed three main types of shoppers. Interestingly, a small group of high-value customers, our VIPs, accounted for a significant portion of overall sales. Seasonal trend analysis revealed that most purchases happened during the final quarter of the year, especially in November. Association rule mining found products that are often bought together. Finally, geographic trends confirmed that while most customers are based in the UK, countries like Germany, France, and Spain also make up valuable markets.

Evaluation

Several limitations and assumptions need to be addressed to contextualize our findings. One key challenge was the lack of predefined product categories, which required more external classification using GPT-3.5. Though helpful, this may have lowered accuracy in the categorization of items especially in seasonal analyses. Additionally, customer clustering was limited to total spending and did not incorporate frequency, recency, or other behavioral metrics which reduces the depth of segmentation. Despite normalization, there were a few extreme

spending values that also impacted the results. In our attempts to analyze relationships between customers and specific products through cross-tabulation, they were hindered by data sparsity which prevented reliable insights from being collected.

Furthermore, the analysis relied on key assumptions such as: that all canceled or returned transactions were correctly removed and that the behavior observed from 2010-2011 transactions reflect broader shopping trends. These constraints can suggest that even though we had effective insights gathered, robust data labeling and expanded behavioral features will be helpful.

Strategic Recommendations

The results of our analysis suggest several strategies for retailers. Retailers should consider developing exclusive loyalty programs and more personalized outreach for customers with high and consistent purchase totals. This can help increase retention and reinforce brand loyalty across their most valuable segments. To encourage greater participation from moderate spenders, businesses should promote “frequently bought together” or bundle complementary products suggestions during checkout. This can not only raise average order value, but introduces customers to new lines of products. For customers with minimal purchasing activity, first-time buyer discounts or targeted campaigns can re-engage their interest and prompt further exploration of the shopping site. Inventory and marketing plans should account for the seasonal demand fluctuations to ramp up campaigns and supply during peak shopping periods. Campaigns that feature gift-oriented or timely promotions can capitalize on high-volume months, especially in the fourth quarter. Lastly, improving product categorization via machine learning models or clearly labeling in data entry can enhance customer-facing recommendations by offering more structure to product groupings.

Future Work

To deepen insights and improve long-term strategy, there are a few areas of further investigation that are recommended. First, analyzing customer lifetime value (CLTV) will help to identify which segments contribute the most to profitability over time, using factors such as frequency and duration of engagement. Predictive modeling can also be used to identify at-risk customers to prevent early signs of churn which will enable proactive retention strategies. Next, product performance analysis including seasonal patterns in sales, returns, and reviews, would allow for improved pricing decisions and inventory. Evaluating marketing effectiveness through metrics such as click-through rates and conversions will provide feedback on the specific campaign’s impact. Finally, examining customer behavior across several channels can reveal cross-selling opportunities and enhance the consistency of customer experience among platforms.

Reference

Chen, D. (2015). Online Retail [Dataset]. UCI Machine Learning Repository.
<https://doi.org/10.24432/C5BW33>.